Research Methods

Day 7 · Academic Writing & Presenting Your Research

July 7, 2026

Welcome to Day 7!

Today is about **communicating your research** — writing it up with academic rigor and presenting it with confidence.

By the end of the morning you will be able to:

- Recognize the **structure of a research paper** — from introduction to references
- Write in a **formal, objective academic style**
- **Cite sources** correctly and know what needs citing
- Draft the **introduction slides** for your team presentation

This week

Schedule for the Week and what we'll be doing

I would like you to send me the literature review today

Intro slide

Tomorrow we will work on your data

Thursday we we will work on the analysis

Friday We will work on the presenting the analysis, methods, and results

Friday select the table, and the figures

So we will have intro, literature review slides ready and most of the paper

Next week

Day 12 July 14 written proposal due and complete slide deck

Day 13 July 15 full reversal of the presentation

Final presentation is on July 17th

Presentation will be 10 to 12 minutes with 3 minutes for questions

The Research Story:
Beginning, Middle, End

The Research Story

Every research paper tells a **story** with the same parts. Each section answers **one question**:

1. **Introduction** — *Why does this matter?*
2. **Literature Review** — *What do we already know?*
3. **Methods** — *How did you study it?*
4. **Results** — *What did you find?*
5. **Discussion** — *What does it mean?*
6. **Conclusion** — *So what? What's next?*
7. **References** — *Whose work did you build on?*

25:00

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This is what you're working toward. We'll look at a real published student research paper — the same arc, written by people your age.

Academic Writing: Style & Standards

Academic Writing Characteristics

Formal Tone

Professional language — no slang or contractions. *"A survey was conducted"*, not *"I asked my friends."*

Objective Focus

Evidence over opinion. *"The data suggest..."* instead of *"I think..."*

Thorough Detail

Enough information for others to replicate your work or evaluate your methods.

Proper Citations

Acknowledge all sources using a standard format — APA, MLA, or Chicago.

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Key principles: one idea per sentence · active voice when possible · precise vocabulary · third person in formal writing · no filler phrases.

Literature Review Template

Begin with a short introduction:

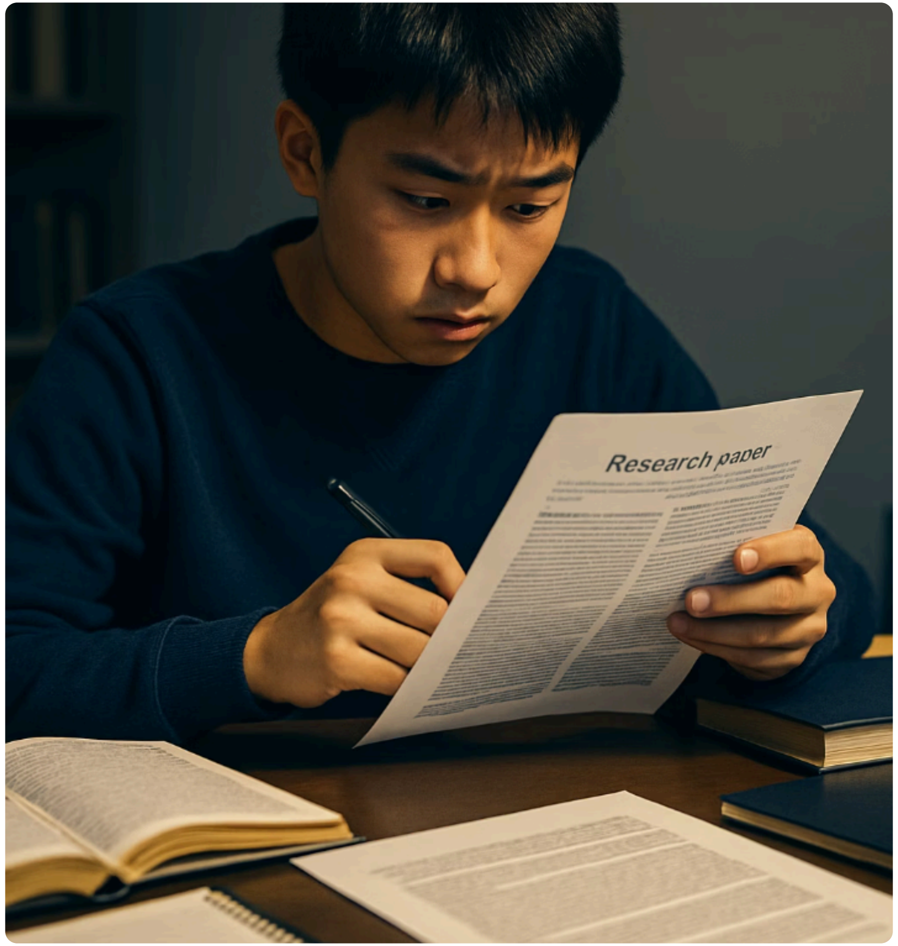
- What is your topic?
- Why is it important?
- Give the broad context of the problem.

Body:

- Share what other studies have found.
- Organize by **themes, not by author**.

End with a summary:

- Summarize what was found.
- Clarify how **your** project connects to past research.
- Name the **patterns** you noticed and the **gaps** that remain.



Practice: Rewrite It in Academic Language

Rewrite each **everyday sentence** into **formal academic language**. Then compare versions as a class.

1. "I think social media is totally messing up how teens sleep."
2. "We asked a bunch of kids and most of them said they were stressed."
3. "Everybody knows that exercise is good for you."
4. "The results were honestly kind of surprising to us."
5. "This study is super important and could change everything."

05:00

Citing Your Sources

Why Citing Matters

What is citing?

Citing your sources means **giving credit** to the original authors of the ideas, words, or data you use in your work.

We cite to:

- **Avoid plagiarism** — taking credit for someone else's work
- **Show where** your information came from
- **Help others find** those sources to learn more

Citing is part of being a trustworthy researcher. It shows respect for others' work — and that you're serious about your own.

What to Cite – and What Not To

✓ You should cite:

- Direct quotes (even short ones)
- Paraphrased ideas (someone else's idea in your own words)
- Statistics or data
- Images, charts, or graphics you didn't create

✕ You don't need to cite:

- Common knowledge (e.g., "Water freezes at 0°C")
- Your own original thoughts

How to Cite

There are different styles (**APA, MLA, Chicago**), but the basic idea is the same: name the **author, date**, and sometimes the **page** in your text — then list the full details at the end.

In-text (APA): "Teens who sleep less are more likely to report symptoms of depression" (Williams, 2021, p. 45).

References page: Williams, A. (2021). *The Science of Teen Sleep.* TeenHealth Publishing.

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For online sources, never paste a bare link — include full source info:

✗ Not this

`https://example.com/article`

✓ Do this

Smith, J. (2022). *Climate change and food security.*

National Environmental Journal.

https://example.com/article

Tips for Citations

Track as you go

Write down author, title, date, and link the moment you find a source.

Don't wait until the end

Build citations in as you write — not the night before it's due.

Use citation tools

EasyBib, Zotero, or the built-in Google Docs citation feature help format correctly.

Double-check the style

Always match your teacher's required style guide.

Building Your Presentation: Introduction Slides

Your Presentation Outline

Your first slides set up the whole story. Each one has a clear job:

Title Slide

Project title, presenters, date, institution. Add logos if you can.

Topic Slide

Name your global challenge, define it, and state why it matters. *Cite your sources.*

Literature Review Slide

A few statements arranged by **themes** — what you know from the literature. *Cite your sources.*

Research Question Slide

Your research question(s) — and a hypothesis if you have one.

Introduction Slides Workshop

Each team drafts its **four introduction slides**.

1. **Title slide** — project, team, date, institution.
2. **Introduction slide** — topic + why it's important + a hook.
3. **Background / Literature slide** — what's known, organized by theme, with citations.
4. **Research Question slide** — your question(s) and hypothesis.

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30:00

Scientific Storytelling:
The 90-Second Intro

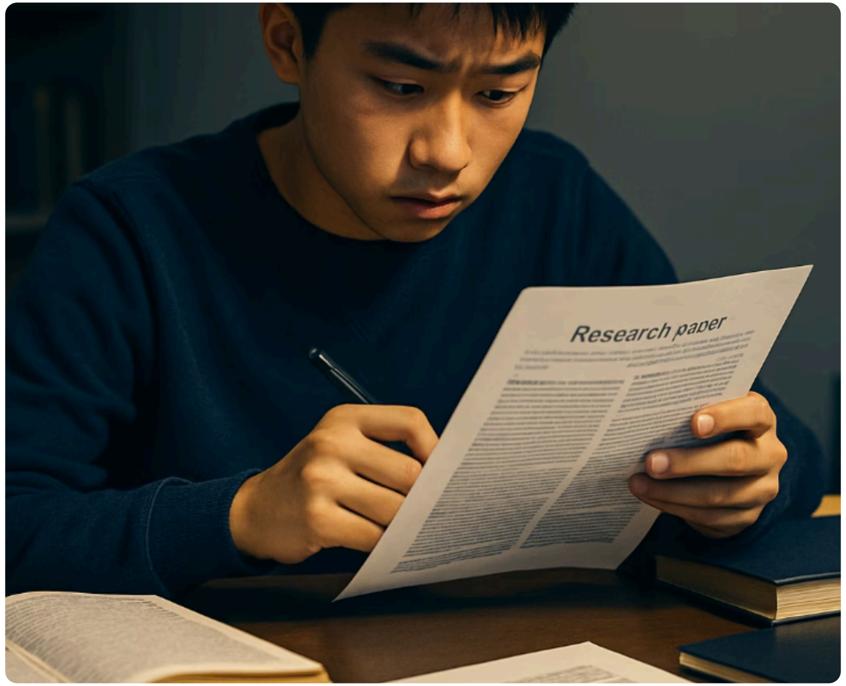
The 90-Second Intro Practice 20 min

Each team presents its **introduction slides** in **90 seconds**.

No reading from the slides — speak *to* the audience.

The class gives **one piece of feedback** per team:

- ⚡ **Most powerful moment**
- 💡 **Needs more clarity**



01:30

Real-World PBL Case: Climate Science 🔥

Does temperature rise affect wildfire frequency?

The data:

- **NOAA** temperature data
- **NIFC** wildfire data

The analysis:

- Run a **correlation:** is average **summer temperature** associated with **annual acres burned**?
- **Plot the trend** over **30 years**.

Interpret critically:

- Is this correlation **causal**?
- What **confounds** exist?
  - Drought
  - Forest management policy
  - Land-use changes

Correlation is not causation — a strong association can hide a third variable driving both.